

Closed All-Orders Endpoint Asymptotics for the Inverse Fabius Function

Diagonal Lagrange–Bürmann reversion, periodic top-jet laws,
phase-locked quantiles, and inverse-derivative hierarchies

Research report based on the complete LaTeX corpus under
`Analysis/FabiusFunction/docs`

Principal contribution. Starting from the repository’s exact-phase forward expansion and its existing recursive inverse engine, this report gives a nonrecursive diagonal Lagrange–Bürmann formula for every inverse coefficient. It proves coefficient-degree and highest-periodic-derivative laws, writes the third inverse correction explicitly, develops an all-order derivative calculus, and verifies the first three corrections on exact dyadic quantiles.

Abstract

Let F be the Fabius distribution function and $G = F^{-1}$. The repository establishes an arbitrary-order small- x expansion of $\log F(x)$ in the exact lower-Lambert phase $\lambda = -W_{-1}(-x \log 2)/\log 2$, with a one-periodic Gamma-zeta fluctuation Ψ and periodic saddle coefficients A_j . A recent inverse-focused draft in the same corpus already derives the first two endpoint corrections for G , their nontrivial phase dependence, and a triangular all-orders recursion.

This report closes the remaining coefficient-formula gap. With

$$T = \log(1/y), \quad \rho = \sqrt{\frac{2T}{\log 2}}, \quad \ell = \log \rho, \quad \theta = \rho + \frac{1}{\log 2} + \frac{1}{2},$$

we prove, for every fixed N , the two-scale expansion

$$G(y) = G_0(y) \left(1 + \sum_{n=1}^N \frac{g_n(\ell, \theta)}{\rho^n} + \mathcal{O}_N \left(\frac{(1+\ell)^{N+1}}{\rho^{N+1}} \right) \right), \quad G_0(y) = \frac{\sqrt{T}}{e\sqrt{\log 2}} e^{-\sqrt{2T} \log 2}.$$

The coefficients are polynomials in ℓ with analytic one-periodic coefficients. A catalytic-variable application of Lagrange-Bürmann inversion gives a finite coefficient extraction formula for every phase coefficient d_n , every logarithmic coefficient h_n , and every multiplicative coefficient g_n . This is a direct formula rather than a step-by-step reversion.

Several structural consequences follow. The universal highest logarithm is $[\ell^n]g_n = 1/(2^n n!)$. The coefficient of the newest periodic derivative is

$$[\Psi^{(2n-2)}]h_n = [\Psi^{(2n-2)}]g_n = \frac{(-1)^n}{2^{n-1}(n-1)!(\log 2)^{2n-2}}.$$

We also obtain an explicit third correction, phase-locked all-orders subsequences, Gamma-zeta convolution formulas for Fourier modes, an all-order recurrence for the quantile elasticity, and for each fixed $m \geq 1$ the two-term derivative law

$$G^{(m)}(y) = (-1)^{m-1}(m-1)! \frac{G(y)}{y^m \rho} \left[1 + \frac{(\Psi'(\theta) - 1)/\log 2 - H_{m-1}}{\rho} + \mathcal{O}_m \left(\frac{1+\ell}{\rho^2} \right) \right].$$

Exact tests at $y_n = F(2^{-n})$ show that the new third correction lowers the relative error to 7.40×10^{-5} at $n = 20$, 6.05×10^{-7} at $n = 80$, and 4.92×10^{-8} at $n = 160$.

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1 Scope, corpus audit, and contribution boundary

1.1 The source corpus

The requested source tree is unusually broad. It contains the primary Fabius–Rvachev exposition, a canonical non-formalized frontier volume, historical versions, a Lean crosswalk, q-series and Thue–Morse studies, papers transcribed into LaTeX, and a rapidly growing set of frontier reports. The inverse-and-sampling report records a recursive audit of 57 LaTeX sources in the pre-consolidation tree. On 28 August 2026 the draft manifest reorganized later arrivals into thematic consolidations: Fourier decay, Thue–Morse, exponent and q-series, spectra and arithmetic, integration and transforms, inverse and sampling, representations, and collected frontiers.

For nonduplication, the audit was claim-oriented rather than merely bibliographic. Every identified source was searched for the endpoint variables, inverse notation, Lambert phase, Ψ , A_j , Bell reversion, Lagrange inversion, derivative asymptotics, and coefficient-structure statements. The endpoint and inverse sources were read line by line; unrelated transform, representation, interpolation, and Fourier-decay sources were read through their abstracts, contents, theorem statements, conclusion, and cross-references. Historical variants were retained in the comparison because some contain formulas absent from the living exposition.

Source family	Material relevant to this report	Audit role
Primary exposition	Exact lower-Lambert phase; $C_{\sharp}$, Ψ ; A_1 , A_2 ; arbitrary-order forward expansion	Analytic input
Canonical frontier volume	Closed saddle jets, Bell recurrences, top-jet law for A_j , explicit higher corrections	Coefficient input
Inverse and Sampling Frontiers	First two inverse corrections, cluster interval, two-term elasticity, triangular all-order engine	Immediate prior art
Archived small-argument studies	Earlier leading equivalents and derivative estimates	Historical boundary
q-series and Thue–Morse studies	Exact dyadic recurrences, Gaussian q-binomial and finite-block identities	Exact test data
Fourier-decay corpus	Infinite sinc products and dyadic spectral scaling	Structural context
Representation and transform volumes	Moment, resolvent, Mellin, sampling, and multiresolution formulas	Cross-connection screening
Embedded papers	Classical definition, arithmetic values, lacunary products, evaluation methods	External prior art

Table 1: Thematic audit of the repository corpus. A path-level ledger is given in [section B](#).

1.2 What was already present

The following are treated as established repository inputs, not as new results of this report:

- the exact Lambert phase and the all-orders forward expansion of $\log F$;
- the Gamma–zeta Fourier coefficients of Ψ and the saddle coefficients A_j ;
- the first two inverse phase corrections d_1 , d_2 and inverse log-corrections h_1 , h_2 ;
- the nondegenerate first normalized cluster interval;
- the triangular Lambert–Bell recursion for arbitrary inverse order;
- the two-term endpoint elasticity $yG'(y)/G(y)$.

1.3 Results added here

New relative to the inspected repository

The main additions are:

1. a diagonal Lagrange–Bürmann coefficient formula for d_n , h_n , and g_n at arbitrary order;
2. a rigorous fixed-order Poincaré theorem with explicit logarithmic remainder grading;
3. a compact explicit third correction d_3, h_3, g_3 and exact symbolic verification;
4. degree laws and closed highest-logarithm formulas, including $[\ell^n]g_n = 1/(2^n n!)$;
5. an inverse top-jet law transferring the forward A_j highest derivative to every inverse coefficient;
6. explicit finite Gamma–zeta convolution formulas for all periodic Fourier modes;
7. phase-locked all-orders quantile expansions and coefficient tomography;
8. an arbitrary-order elasticity recurrence and a two-term formula for every fixed inverse derivative $G^{(m)}$.

Priority statement

“New” means absent from the inspected repository sources under the normalizations used there. The targeted literature screening found no indexed external treatment of these exact inverse endpoint coefficients, but this report does not claim a complete worldwide priority search across every normalization of the Fabius or Rvachev functions.

2 Normalization and the exact-phase forward input

2.1 Fabius and inverse Fabius functions

We use the standard distribution normalization. The Fabius function $F : [0, 1] \rightarrow [0, 1]$ is the strictly increasing C^∞ function characterized on the left half by

$$F'(x) = 2F(2x), \quad 0 \leq x \leq \frac{1}{2}, \quad (2.1)$$

with $F(0) = 0$, $F(1) = 1$, and reflection $F(x) + F(1 - x) = 1$. Its inverse is

$$G : (0, 1) \longrightarrow (0, 1), \quad F(G(y)) = y. \quad (2.2)$$

The present report concerns $y \downarrow 0$. Reflection gives the right-endpoint formulas automatically:

$$G(1 - y) = 1 - G(y). \quad (2.3)$$

Put

$$L = \log 2, \quad B = 1 + \frac{L}{2}, \quad \beta = \frac{B}{L} = \frac{1}{L} + \frac{1}{2}. \quad (2.4)$$

For $x > 0$ sufficiently small, the exact lower-Lambert phase is

$$\lambda(x) = -\frac{1}{L}W_{-1}(-Lx), \quad x = \lambda(x)e^{-L\lambda(x)}. \quad (2.5)$$

The identity on the right is exact, not asymptotic.

2.2 Gamma–zeta periodicity

Define

$$\chi_k = \frac{2\pi i k}{L}, \quad \widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L} \quad (k \in \mathbb{Z} \setminus \{0\}), \quad (2.6)$$

with $\widehat{\Psi}(0) = 0$, and

$$\Psi(u) = \sum_{k \neq 0} \widehat{\Psi}(k) e^{2\pi i k u}. \quad (2.7)$$

Then Ψ is real-valued, smooth, one-periodic, and has mean zero. The constant in the sharp endpoint expansion is

$$C_{\sharp} = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi, \quad (2.8)$$

where γ and γ_1 are the Euler and first Stieltjes constants.

2.3 All-orders forward theorem

The analytic input is the repository's exact-phase expansion: for every fixed $N \geq 1$,

$$\begin{aligned} \log F(x) = & -\frac{L}{2}\lambda^2 + B\lambda - \frac{1}{2}\log \lambda + C_{\sharp} + \Psi(\lambda) \\ & + \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}_N(\lambda^{-N}), \quad \lambda = \lambda(x) \rightarrow \infty. \end{aligned} \quad (2.9)$$

Every A_j is a bounded smooth one-periodic differential polynomial in Ψ . The first coefficient is

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \quad (2.10)$$

The forward top-jet theorem states

$$[\Psi^{(2j)}]A_j = \frac{(-1)^j}{2^j j! L^{2j}}. \quad (2.11)$$

This exceptionally simple coefficient will propagate to the inverse expansion in [section 8.2](#).

3 The two-scale inverse problem

For $y \downarrow 0$, set

$$T = \log \frac{1}{y}, \quad \rho = \sqrt{\frac{2T}{L}}, \quad z = \rho^{-1}, \quad \ell = \log \rho, \quad \theta = \rho + \beta. \quad (3.1)$$

The leading inverse factor is

$$G_0(y) = \rho e^{-L\rho - L\beta} = \frac{\sqrt{T}}{e\sqrt{L}} e^{-\sqrt{2LT}}. \quad (3.2)$$

Write the exact phase of the true quantile as

$$\lambda(G(y)) = \rho + \beta + D, \quad D = D(z; \ell, \theta). \quad (3.3)$$

The key point is that z is the small variable while ℓ and θ must remain coefficient variables. Replacing θ by a constant or by a truncated elementary approximation destroys terms of the same order as the desired correction.

The inverse-focused repository draft obtains

$$D = \frac{d_1}{\rho} + \frac{d_2}{\rho^2} + \mathcal{O}\left(\frac{\ell^2}{\rho^3}\right), \quad (3.4)$$

where

$$d_1 = \frac{\beta^2}{2} + \frac{C_\# + \Psi(\theta) - \frac{1}{2}\ell}{L}, \quad (3.5)$$

$$d_2 = \frac{\Psi'(\theta)d_1 + A_1(\theta) - \beta/2}{L}. \quad (3.6)$$

Because $G/G_0 = (1 + \beta z + zD)e^{-LD}$, this gives

$$\log \frac{G}{G_0} = \frac{h_1}{\rho} + \frac{h_2}{\rho^2} + \mathcal{O}\left(\frac{\ell^3}{\rho^3}\right), \quad (3.7)$$

with

$$h_1 = \frac{1}{2}\ell + \kappa_0 - \Psi(\theta), \quad \kappa_0 = \beta - \frac{L}{2}\beta^2 - C_\#, \quad (3.8)$$

$$h_2 = (1 - \Psi'(\theta))d_1 - \frac{\beta^2}{2} - A_1(\theta) + \frac{\beta}{2}. \quad (3.9)$$

The periodic term in h_1 implies that no phase-blind constant can replace the first normalized correction. The goal now is to turn the recursive continuation of these formulas into a direct all-orders coefficient calculus.

4 Formal endpoint equation and coefficient algebra

4.1 Reduction to a single implicit series

Put $\lambda = \rho + \beta + D$ in [equation \(2.9\)](#). Since $B = L\beta$ and $T = L\rho^2/2$, the quadratic part collapses exactly:

$$-\frac{L}{2}\lambda^2 + B\lambda = -T - L\rho D + \frac{L}{2}\beta^2 - \frac{L}{2}D^2. \quad (4.1)$$

Moreover

$$\log \lambda = \ell + \log(1 + \beta z + zD), \quad \lambda^{-j} = z^j(1 + \beta z + zD)^{-j}. \quad (4.2)$$

After equating $\log F(G(y)) = -T$, the inverse problem becomes

$$0 = \frac{L}{z}D + \mathcal{R}(z, D), \quad (4.3)$$

where, at arbitrary formal order,

$$\begin{aligned} \mathcal{R}(z, w) = & -\frac{L}{2}\beta^2 + \frac{L}{2}w^2 + \frac{1}{2}\ell + \frac{1}{2}\log(1 + \beta z + zw) - C_\# - \Psi(\theta + w) \\ & - \sum_{j \geq 1} A_j(\theta + w)z^j(1 + \beta z + zw)^{-j}. \end{aligned} \quad (4.4)$$

The sum is formal in z ; to compute through z^N , only $A_1, \dots, A_{N-1}$ are needed. Define

$$\begin{aligned} \Phi(z, w) := & -\frac{1}{L}\mathcal{R}(z, w) = \frac{\beta^2}{2} - \frac{w^2}{2} - \frac{\ell}{2L} - \frac{1}{2L}\log(1 + \beta z + zw) + \frac{C_\# + \Psi(\theta + w)}{L} \\ & + \frac{1}{L} \sum_{j \geq 1} A_j(\theta + w)z^j(1 + \beta z + zw)^{-j}. \end{aligned} \quad (4.5)$$

Then the endpoint inversion is the compact fixed-point equation

$$\boxed{D = z\Phi(z, D)}. \quad (4.6)$$

This form isolates the large derivative L/z in [equation \(4.3\)](#) and exposes an ordinary Lagrange inversion problem.

4.2 The coefficient ring

Let $\mathcal{A}$ be the differential algebra of one-periodic functions generated by Ψ and the forward coefficients A_j . The Gamma–zeta decay reviewed in [section 9](#) implies that its elements are real analytic in θ . We work in

$$\mathcal{R} = \mathcal{A}[\ell][[z]], \quad (4.7)$$

where differentiation with respect to θ acts on $\mathcal{A}$ and ℓ is an algebraically independent slow variable. Taylor expansions such as

$$\Psi(\theta + w) = \sum_{r \geq 0} \frac{\Psi^{(r)}(\theta)}{r!} w^r \quad (4.8)$$

are interpreted in $\mathcal{R}[[w]]$. At a fixed order n , every coefficient extraction below uses only finitely many Taylor jets and finitely many A_j .

Remark 4.1 (Two scales are indispensable). The variable $\ell = \log \rho$ is unbounded but grows more slowly than every positive power of ρ , whereas $\theta = \rho + \beta$ winds around the unit circle. Thus the expansion is Poincaré in $z = \rho^{-1}$ with polynomial-logarithmic, periodically modulated coefficients. It is not a power series with numerical constants.

5 A complete fixed-order asymptotic theorem

Theorem 5.1 (All-orders inverse endpoint expansion). *For every fixed integer $N \geq 1$, there exist uniquely determined coefficients*

$$d_n(\ell, \theta), \quad h_n(\ell, \theta), \quad g_n(\ell, \theta) \quad (1 \leq n \leq N), \quad (5.1)$$

which are polynomials in ℓ with real-analytic one-periodic coefficients in θ , such that, as $y \downarrow 0$,

$$\lambda(G(y)) = \rho + \beta + \sum_{n=1}^N \frac{d_n(\ell, \theta)}{\rho^n} + \mathcal{O}_N\left(\frac{(1+\ell)^N}{\rho^{N+1}}\right), \quad (5.2)$$

$$\log \frac{G(y)}{G_0(y)} = \sum_{n=1}^N \frac{h_n(\ell, \theta)}{\rho^n} + \mathcal{O}_N\left(\frac{(1+\ell)^N}{\rho^{N+1}}\right), \quad (5.3)$$

$$\frac{G(y)}{G_0(y)} = 1 + \sum_{n=1}^N \frac{g_n(\ell, \theta)}{\rho^n} + \mathcal{O}_N\left(\frac{(1+\ell)^{N+1}}{\rho^{N+1}}\right). \quad (5.4)$$

The estimates are uniform in the phase $\theta \bmod 1$. The coefficient d_n depends only on $A_1, \dots, A_{n-1}$ and on finitely many derivatives of those functions and of Ψ . General finite coefficient-extraction formulas are given in [theorem 6.1](#).

Proof. The all-orders forward theorem, truncated after A_{N-1} , has a remainder $\mathcal{O}_N(\lambda^{-N}) = \mathcal{O}_N(z^N)$ uniformly in the real phase. The leading inverse estimate already present in the corpus gives $D = \mathcal{O}(\ell z)$. Substitute a formal polynomial

$$D^{[N]} = \sum_{n=1}^N d_n z^n \quad (5.5)$$

into [equation \(4.6\)](#) and choose the coefficients so that all powers $z, \dots, z^N$ cancel. Triangularity follows because D is multiplied by z : the coefficient of z^n on the right depends only on $d_1, \dots, d_{n-1}$. Hence d_n exists uniquely. Since the sole explicit occurrence of ℓ in Φ is affine, induction shows polynomial dependence on ℓ ; periodic analyticity follows from closure of $\mathcal{A}$ under products, differentiation, and composition with a formal increment.

The residual in [equation \(4.3\)](#) at $D^{[N]}$ is $\mathcal{O}_N((1+\ell)^N z^N)$. On the segment joining $D^{[N]}$ to the exact D ,

$$\partial_w \left(\frac{L}{z} w + \mathcal{R}(z, w) \right) = \frac{L}{z} + \mathcal{O}_N(1), \quad (5.6)$$

where the error is uniform in phase. The mean-value theorem therefore gains one power of z and proves [equation \(5.2\)](#).

Finally the exact Lambert identity $x = \lambda e^{-L\lambda}$ gives

$$\frac{G(y)}{G_0(y)} = (1 + \beta z + zD)e^{-LD}, \quad (5.7)$$

so that

$$H(z) := \log \frac{G}{G_0} = \log(1 + \beta z + zD) - LD. \quad (5.8)$$

Formal composition gives [equation \(5.3\)](#); exponentiation gives [equation \(5.4\)](#). The extra displayed power of $(1 + \ell)$ in the last remainder is a harmless uniform bound for the first omitted complete Bell polynomial. $\square$

Corollary 5.2 (Right endpoint). *Since $F(x) + F(1 - x) = 1$, one has $G(1 - y) = 1 - G(y)$. Thus every formula in [theorem 5.1](#) immediately supplies the complete expansion at $y \uparrow 1$ by replacing $G(y)$ with $1 - G(1 - y)$.*

6 Diagonal Lagrange–Bürmann coefficient formulas

The repository's triangular recursion is efficient, but it still computes order n only after all lower orders have been substituted. A catalytic variable turns the same fixed-point equation into a closed finite extraction at any prescribed order.

Theorem 6.1 (Diagonal phase, logarithmic, and multiplicative formulas). *Let Φ be defined by [equation \(4.5\)](#). Then*

$$d_n = \sum_{k=1}^n \frac{1}{k} [z^{n-k} w^{k-1}] \Phi(z, w)^k. \quad (6.1)$$

More generally, for any formal $K(z, w)$,

$$\begin{aligned} [z^n] K(z, D(z)) &= [z^n] K(z, 0) \\ &+ \sum_{k=1}^n \frac{1}{k} [z^{n-k} w^{k-1}] \partial_w K(z, w) \Phi(z, w)^k. \end{aligned} \quad (6.2)$$

Consequently, with

$$H(z, w) = \log(1 + \beta z + zw) - Lw, \quad (6.3)$$

$$M(z, w) = (1 + \beta z + zw)e^{-Lw}, \quad (6.4)$$

we have

$$h_n = \frac{(-1)^{n+1} \beta^n}{n} + \sum_{k=1}^n \frac{1}{k} [z^{n-k} w^{k-1}] \left(\frac{z}{1 + \beta z + zw} - L \right) \Phi(z, w)^k, \quad (6.5)$$

$$g_n = \beta \mathbf{1}_{\{n=1\}} + \sum_{k=1}^n \frac{1}{k} [z^{n-k} w^{k-1}] e^{-Lw} \{z - L(1 + \beta z + zw)\} \Phi(z, w)^k. \quad (6.6)$$

Every sum is finite, and only the z^{n-1} truncation of Φ is required.

Proof. Introduce a catalytic variable ξ and solve

$$D(\xi, z) = \xi \Phi(z, D(\xi, z)). \quad (6.7)$$

For fixed z , ordinary Lagrange–Bürmann inversion in ξ gives

$$[\xi^k]D(\xi, z) = \frac{1}{k}[w^{k-1}]\Phi(z, w)^k. \quad (6.8)$$

Set $\xi = z$ and extract total degree n ; this gives [equation \(6.1\)](#). The composition form of Lagrange–Bürmann gives

$$K(z, D(\xi, z)) = K(z, 0) + \sum_{k \geq 1} \frac{\xi^k}{k} [w^{k-1}] \partial_w K(z, w) \Phi(z, w)^k. \quad (6.9)$$

Again set $\xi = z$ and extract z^n , proving [equation \(6.2\)](#). The specializations follow from

$$\partial_w H(z, w) = \frac{z}{1 + \beta z + zw} - L, \quad (6.10)$$

$$\partial_w M(z, w) = e^{-Lw} \{z - L(1 + \beta z + zw)\}, \quad (6.11)$$

and from $H(z, 0) = \log(1 + \beta z)$ and $M(z, 0) = 1 + \beta z$. $\square$

Corollary 6.2 (Bell-polynomial conversion). *The two coefficient normalizations are related by the complete exponential Bell polynomials:*

$$g_n = \frac{1}{n!} \mathcal{B}_n(1!h_1, 2!h_2, \dots, n!h_n). \quad (6.12)$$

Equivalently, $g_0 = 1$ and

$$ng_n = \sum_{j=1}^n j h_j g_{n-j}. \quad (6.13)$$

Algorithm 6.3 (Arbitrary-order symbolic generation). To compute the inverse expansion through order N :

1. obtain $A_1, \dots, A_{N-1}$ from the forward saddle calculus;
2. Taylor-expand $\Psi(\theta + w)$ and $A_j(\theta + w)$ only as far as the bidegrees in [equation \(6.1\)](#) require;
3. truncate Φ at z^{N-1} ;
4. evaluate the finite diagonal extractions [equation \(6.1\)](#);
5. use either [equations \(6.5\)](#) and [\(6.6\)](#) directly or the Bell conversion [equation \(6.12\)](#).

For a single high order, this avoids constructing every intermediate full composition. For all orders up to N , the triangular and diagonal algorithms are complementary: the former reuses lower-order work, while the latter gives an independent certificate.

7 The explicit third correction

Write $P_r = \Psi^{(r)}(\theta)$. Besides [equation \(2.10\)](#), the third inverse order requires A_2 . In the repository normalization it is

$$A_2 = \frac{P_1}{24L} + \frac{2P_1 + 1}{4L^2} - \frac{P_1^3 + 3P_1^2 + 3P_1P_2 + 3P_2 + P_3}{6L^3} + \frac{4P_1^2P_2 + 4P_1P_3 + 2P_2^2 + P_4}{8L^4}. \quad (7.1)$$

Also

$$A'_1 = -\frac{P_3 + 2P_1P_2}{2L^2}. \quad (7.2)$$

Theorem 7.1 (Closed third-order inverse coefficients). *With d_1, d_2 as in equations (3.5) and (3.6), the third phase coefficient is*

$$d_3 = -\frac{1}{L} \left\{ \frac{L}{2} d_1^2 + \frac{1}{2} d_1 - \frac{\beta^2}{4} - P_1 d_2 - \frac{P_2}{2} d_1^2 - A_1' d_1 + \beta A_1 - A_2 \right\}. \quad (7.3)$$

The logarithmic coefficients through third order are

$$h_1 = \beta - L d_1, \quad (7.4)$$

$$h_2 = d_1 - \frac{\beta^2}{2} - L d_2, \quad (7.5)$$

$$h_3 = d_2 - \beta d_1 + \frac{\beta^3}{3} - L d_3. \quad (7.6)$$

The corresponding multiplicative coefficients are

$$g_1 = h_1, \quad (7.7)$$

$$g_2 = h_2 + \frac{1}{2} h_1^2, \quad (7.8)$$

$$g_3 = h_3 + h_1 h_2 + \frac{1}{6} h_1^3. \quad (7.9)$$

Proof. Expand equation (4.3) after inserting $D = d_1 z + d_2 z^2 + d_3 z^3 + \mathcal{O}(z^4)$. The coefficients of z^0 and z^1 reproduce d_1 and d_2 . The coefficient of z^2 is

$$L d_3 + \frac{L}{2} d_1^2 + \frac{1}{2} d_1 - \frac{\beta^2}{4} - P_1 d_2 - \frac{P_2}{2} d_1^2 - A_1' d_1 + \beta A_1 - A_2, \quad (7.10)$$

which proves equation (7.3). Expanding equation (5.8) gives equations (7.4) to (7.6); exponentiation gives equation (7.9). $\square$

Remark 7.2 (Why the compact form matters). Fully expanding d_3 into $\ell, P_0, \dots, P_4$ is possible but obscures its origin. Formula equation (7.3) displays five mechanisms separately: the quadratic saddle correction, the logarithmic Jacobian, Taylor transport of Ψ , transport of A_1 , and the genuinely new forward coefficient A_2 . It is also substantially more stable for symbolic and numerical evaluation.

8 Coefficient structure: logarithmic degree and universal laws

The diagonal formula does more than compute coefficients: its bidegree geometry reveals features that are not apparent in a triangular expansion.

8.1 Highest powers of the slow logarithm

Set

$$a = -\frac{1}{2L}, \quad \phi(w) = \frac{\beta^2}{2} + \frac{C_\#}{L} + \frac{\Psi(\theta + w)}{L} - \frac{w^2}{2}. \quad (8.1)$$

Then the z^0 slice of equation (4.5) is

$$\Phi(0, w) = a\ell + \phi(w). \quad (8.2)$$

Theorem 8.1 (Logarithmic degree and top-log coefficient). *The phase coefficients satisfy*

$$\deg_\ell d_1 = 1, \quad \deg_\ell d_n \leq n - 1 \quad (n \geq 2), \quad (8.3)$$

and, for $n \geq 2$,

$$[\ell^{n-1}]d_n = a^{n-1}[w^{n-1}]\phi(w). \quad (8.4)$$

In particular,

$$[\ell]d_2 = -\frac{\Psi'(\theta)}{2L^2}, \quad (8.5)$$

$$[\ell^2]d_3 = \frac{\Psi''(\theta) - L}{8L^3}, \quad (8.6)$$

and for $n \geq 4$,

$$[\ell^{n-1}]d_n = \frac{(-1)^{n-1}\Psi^{(n-1)}(\theta)}{2^{n-1}L^n(n-1)!}. \quad (8.7)$$

The logarithmic coefficients obey

$$\deg_\ell h_1 = \deg_\ell h_2 = 1, \quad \deg_\ell h_n \leq n-1 \quad (n \geq 3), \quad (8.8)$$

with

$$[\ell]h_1 = \frac{1}{2}, \quad (8.9)$$

$$[\ell]h_2 = \frac{\Psi'(\theta) - 1}{2L}, \quad (8.10)$$

$$[\ell^2]h_3 = \frac{L - \Psi''(\theta)}{8L^2}, \quad (8.11)$$

and, for $n \geq 4$,

$$[\ell^{n-1}]h_n = \frac{(-1)^n \Psi^{(n-1)}(\theta)}{2^{n-1}L^{n-1}(n-1)!}. \quad (8.12)$$

Finally,

$$\deg_\ell g_n \leq n, \quad [\ell^n]g_n = \frac{1}{2^n n!}. \quad (8.13)$$

The highest logarithm of the multiplicative expansion is universal: it contains no periodic data.

Proof. In equation (6.1), a product Φ^k has ℓ -degree at most k . If $k < n$, this is at most $n-1$. For $k = n$, the requested coefficient is $[w^{n-1}]\Phi(0, w)^n/n$. Its nominal ℓ^n term has no w and therefore vanishes for $n \geq 2$. To obtain $\ell^{n-1}w^{n-1}$, choose $a\ell$ from exactly $n-1$ factors and $\phi(w)$ from one factor. The factor n cancels $1/n$, proving equation (8.4). Expanding equation (8.1) gives the displayed special cases.

In $H = \log(1 + \beta z + zD) - LD$, the logarithm at order z^n contains only $d_1, \dots, d_{n-1}$. For $n \geq 3$ its ℓ -degree is at most $n-2$, so the highest logarithm comes from $-Ld_n$. The exceptional orders $n = 1, 2$ follow directly from equations (7.4) and (7.5).

For g_n , use $\exp(H)$. The only way to attain both total z -weight n and ℓ -degree n is to choose $(h_1 z)^n/n!$ and within each h_1 choose its term $\ell/2$. Every h_j with $j \geq 2$ has at least one unit of logarithmic-degree deficit. This proves equation (8.13). $\square$

Corollary 8.2 (The first subleading multiplicative logarithm). Write

$$h_1 = \frac{1}{2}\ell + q(\theta), \quad q(\theta) = \kappa_0 - \Psi(\theta), \quad (8.14)$$

and put $\alpha_j(\theta) = [\ell^{j-1}]h_j$ for $j \geq 2$. Then, for every $n \geq 2$,

$$[\ell^{n-1}]g_n = \frac{q(\theta)}{2^{n-1}(n-1)!} + \sum_{j=2}^n \frac{\alpha_j(\theta)}{2^{n-j}(n-j)!}. \quad (8.15)$$

Thus the first nonuniversal logarithmic layer is still available in closed form from the top-log sequence of h_j .

Proof. In a Bell monomial contributing to g_n , the logarithmic degree is smaller than the z -weight by one precisely when either one factor h_1 contributes q instead of $\ell/2$, or one factor h_j with $j \geq 2$ contributes its top logarithm $\alpha_j \ell^{j-1}$. Two such choices would produce deficit at least two. Counting the remaining copies of h_1 gives [equation \(8.15\)](#). $\square$

8.2 Transfer of the newest periodic derivative

The forward top-jet law [equation \(2.11\)](#) has a clean inverse counterpart.

Theorem 8.3 (Inverse top-jet law). *Regard every coefficient as a differential polynomial in Ψ . For every $n \geq 1$,*

$$[\Psi^{(2n-2)}]d_n = \frac{(-1)^{n-1}}{2^{n-1}(n-1)!L^{2n-1}}, \quad (8.16)$$

$$[\Psi^{(2n-2)}]h_n = [\Psi^{(2n-2)}]g_n = \frac{(-1)^n}{2^{n-1}(n-1)!L^{2n-2}}. \quad (8.17)$$

Here $[\Psi^{(r)}]$ means the coefficient of the monomial that is linear in $\Psi^{(r)}$ and contains no other Ψ -jet factor.

Proof. For $n = 1$, the claims are immediate from $d_1 = \Psi/L + \dots$ and $h_1 = g_1 = -\Psi + \dots$. Let $n \geq 2$. In [equation \(6.1\)](#), the $k = 1$ term contains

$$\frac{1}{L} A_{n-1}(\theta) z^{n-1} \quad (8.18)$$

inside Φ . By [equation \(2.11\)](#), this contributes the right side of [equation \(8.16\)](#).

It is the unique source of derivative order $2n - 2$. Indeed, if a factor $A_j(\theta + w)z^j$ occurs in a term with Lagrange index k , then $j \leq n - k$ and at most $k - 1$ derivatives can be supplied by the w extraction. Its largest possible jet order is therefore

$$2j + (k - 1) \leq 2(n - k) + k - 1 = 2n - k - 1, \quad (8.19)$$

which reaches $2n - 2$ only for $k = 1$ and $j = n - 1$. Pure Ψ factors have lower derivative order under the same bidegree constraint.

At order n , the logarithm in [equation \(5.8\)](#) uses only $d_1, \dots, d_{n-1}$ and therefore has jets of order at most $2n - 4$. Hence the new jet in h_n is $-L$ times that in d_n . Bell products involving h_j with $j < n$ again have jet order at most $2n - 4$, so g_n has the same newest-jet coefficient as h_n . $\square$

Remark 8.4 (A triangular fingerprint). At each inverse order a genuinely new derivative $\Psi^{(2n-2)}$ enters linearly and with a nonzero universal coefficient. Consequently the periodic hierarchy does not close at any finite jet order. This gives a simple diagnostic for computer algebra: omission or an incorrect sign in A_{n-1} is immediately visible in h_n and g_n .

9 Fourier-mode algebra and analytic periodic coefficients

The fluctuation has the absolutely convergent Fourier series

$$\Psi(u) = \sum_{k \in \mathbb{Z} \setminus \{0\}} \widehat{\Psi}(k) e^{2\pi i k u}, \quad \widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k) \zeta(1 - \chi_k)}{L}, \quad \chi_k = \frac{2\pi i k}{L}. \quad (9.1)$$

The zero mode is included in $C_\sharp$, so $\widehat{\Psi}(0) = 0$ in this normalization.

Proposition 9.1 (Finite Gamma–zeta convolution formula). *Let*

$$\mathcal{M}(u) = \prod_{r \geq 0} (\Psi^{(r)}(u))^{m_r} \quad (9.2)$$

be a finite jet monomial. Its Fourier coefficient is

$$\widehat{\mathcal{M}}(k) = \sum_{\substack{k_{r,s} \in \mathbb{Z} \setminus \{0\} \\ \sum_{r,s} k_{r,s} = k}} \prod_{r \geq 0} \prod_{s=1}^{m_r} (2\pi i k_{r,s})^r \widehat{\Psi}(k_{r,s}). \quad (9.3)$$

Every Fourier coefficient of every d_n, h_n, g_n is a finite linear combination of expressions of the form equation (9.3), with coefficients polynomial in ℓ and rational in L and β .

Proof. Differentiate equation (2.7) termwise and multiply the resulting absolutely convergent series. The coefficient of total frequency k is the convolution equation (9.3). The diagonal formulas use finite sums, products, Taylor derivatives, and the forward differential polynomials A_j , so no other periodic operation occurs at a fixed order. $\square$

Proposition 9.2 (Common strip of analyticity). *For every $\sigma < \pi/(2L)$, the functions Ψ, A_j , and all periodic coefficient functions appearing in d_n, h_n, g_n extend holomorphically and boundedly to the closed strip*

$$|\Im u| \leq \sigma. \quad (9.4)$$

Their Fourier series and all differentiated series converge normally there.

Proof. Stirling’s formula on vertical lines gives, for every $\varepsilon > 0$,

$$|\widehat{\Psi}(k)| \leq C_\varepsilon \exp \left[- \left(\frac{\pi^2}{L} - \varepsilon \right) |k| \right]. \quad (9.5)$$

Polynomial factors from differentiation do not change the exponential type. Multiplication corresponds to convolution and preserves normal convergence on compact sub-strips. Since $|e^{2\pi i k u}| = e^{-2\pi k \Im u}$, the limiting half-width is $(\pi^2/L)/(2\pi) = \pi/(2L)$. $\square$

Corollary 9.3 (Certified Fourier truncation). *For fixed inverse order N , phase derivatives up to any fixed order, and $|\Im \theta| \leq \sigma < \pi/(2L)$, truncating the Fourier sums at $|k| \leq K$ incurs an error bounded by*

$$C_{N,\sigma,r} (1 + \ell)^N K^{r+C_N} e^{-(\pi^2/L - 2\pi\sigma)K}. \quad (9.6)$$

Thus only a small number of modes is needed for high-precision real-phase evaluation.

10 Phase locking and all-orders coefficient tomography

The periodic variable can be frozen exactly while the endpoint is approached. This separates the slow logarithm from the dyadic phase and turns the formal coefficient functions into observable limits.

Theorem 10.1 (Phase-locked endpoint sequences). *Fix $u \in [0, 1)$ and let*

$$\rho_{m,u} = m + u - \beta, \quad y_{m,u} = \exp \left(-\frac{L}{2} \rho_{m,u}^2 \right), \quad (10.1)$$

for integers m large enough that $\rho_{m,u} > 0$. Then $\theta = m + u \equiv u \pmod{1}$, and for every fixed N ,

$$\frac{G(y_{m,u})}{G_0(y_{m,u})} = 1 + \sum_{n=1}^N \frac{g_n(\log \rho_{m,u}, u)}{\rho_{m,u}^n} + \mathcal{O}_N \left(\frac{(1 + \log \rho_{m,u})^{N+1}}{\rho_{m,u}^{N+1}} \right), \quad (10.2)$$

where the constant is uniform for $u \in [0, 1]$.

Proof. The definitions give $T = L\rho_{m,u}^2/2$ and $\theta = \rho_{m,u} + \beta = m + u$. Apply [theorem 5.1](#); uniformity follows from compactness and periodic analyticity. $\square$

Corollary 10.2 (First normalized cluster set). *As $y \downarrow 0$, the set of limit points of*

$$\rho \left(\frac{G(y)}{G_0(y)} - 1 \right) - \frac{1}{2}\ell \quad (10.3)$$

contains, and by the first-order expansion equals,

$$\left[\kappa_0 - \max_{u \in [0,1]} \Psi(u), \kappa_0 - \min_{u \in [0,1]} \Psi(u) \right]. \quad (10.4)$$

Every point is realized by a phase-locked sequence.

10.1 Higher coefficient tomography

Write

$$g_n(\ell, u) = \sum_{r=0}^n g_{n,r}(u) \ell^r. \quad (10.5)$$

After subtracting the first $N - 1$ inverse orders, define

$$\mathcal{T}_{N,m}(u) = \rho_{m,u}^N \left\{ \frac{G(y_{m,u})}{G_0(y_{m,u})} - 1 - \sum_{n=1}^{N-1} \frac{g_n(\log \rho_{m,u}, u)}{\rho_{m,u}^n} \right\}. \quad (10.6)$$

Then

$$\mathcal{T}_{N,m}(u) = g_N(\log \rho_{m,u}, u) + \mathcal{O}_N \left(\frac{(1 + \log \rho_{m,u})^{N+1}}{\rho_{m,u}} \right). \quad (10.7)$$

Thus, for each fixed phase, the periodic functions $g_{N,r}(u)$ can in principle be reconstructed from a polynomial fit in $\log \rho$ after lower orders are removed. The universal value $g_{N,N} = 1/(2^N N!)$ supplies an exact normalization and reduces the number of unknown periodic layers by one.

Interpretation

The endpoint does not have a single asymptotic constant after the elementary factor is removed. It has a hierarchy of periodic coefficient profiles. Phase locking converts each profile into a one-parameter family of asymptotic experiments, while [equation \(9.3\)](#) predicts its Fourier spectrum explicitly.

11 Elasticity and every fixed inverse derivative

Inverse derivatives are singular at the endpoint, but the logarithmic quantile expansion contains enough information to organize them to arbitrary order. The correct differential operator must act simultaneously on the fast phase, the slow logarithm, and the small parameter.

11.1 Endpoint differential operator

Since $T = L\rho^2/2$ and $y = e^{-T}$,

$$\mathcal{Y} := y \frac{d}{dy} = -\frac{d}{dT} = -\frac{z}{L} \frac{d}{d\rho}. \quad (11.1)$$

On a coefficient $f(z, \ell, \theta)$ the total ρ derivative is

$$\mathfrak{d} := \partial_\theta + z\partial_\ell - z^2\partial_z, \quad \mathcal{Y} = -\frac{z}{L}\mathfrak{d}. \quad (11.2)$$

Let

$$\mathcal{E}(y) := \frac{yG'(y)}{G(y)} = \mathcal{Y} \log G(y) \quad (11.3)$$

be the endpoint elasticity. Since $\log G_0 = \ell - L\rho - L\beta$, [equation \(5.8\)](#) gives the exact formal identity

$$\boxed{\mathcal{E} = z - \frac{z^2}{L} - \frac{z}{L} \mathfrak{d}H.} \quad (11.4)$$

Theorem 11.1 (All-orders elasticity coefficients). *Write*

$$H = \sum_{n \geq 1} h_n z^n, \quad \mathcal{E} = \sum_{m \geq 1} e_m z^m. \quad (11.5)$$

Then $e_1 = 1$ and, for $m \geq 2$,

$$\boxed{e_m = -\frac{\delta_{m,2}}{L} - \frac{1}{L} \{ \partial_\theta h_{m-1} + \partial_\ell h_{m-2} - (m-2)h_{m-2} \},} \quad (11.6)$$

where $h_0 := 0$. In particular,

$$\mathcal{E} = \frac{1}{\rho} + \frac{\Psi'(\theta) - 1}{L\rho^2} + \mathcal{O}\left(\frac{1+\ell}{\rho^3}\right). \quad (11.7)$$

Thus the complete expansion of G immediately gives the complete expansion of its logarithmic derivative without a new inversion.

Proof. For a single term $h_n z^n$,

$$\mathfrak{d}(h_n z^n) = (\partial_\theta h_n) z^n + (\partial_\ell h_n - n h_n) z^{n+1}. \quad (11.8)$$

Insert this into [equation \(11.4\)](#) and collect z^m . Since $\partial_\theta h_1 = -\Psi'(\theta)$, the special case $m = 2$ gives [equation \(11.7\)](#). $\square$

11.2 Higher ordinary derivatives

For $m \geq 0$, define the dimensionless derivative ratios

$$Q_m(y) = \frac{y^m G^{(m)}(y)}{G(y)}, \quad Q_0 = 1. \quad (11.9)$$

A direct differentiation gives an exact nonlinear-linear recurrence:

$$\boxed{Q_{m+1} = \mathcal{Y} Q_m + (\mathcal{E} - m) Q_m.} \quad (11.10)$$

It is nonlinear only through the already known coefficient series $\mathcal{E}$; for fixed $\mathcal{E}$ it is linear in Q_m .

Theorem 11.2 (All-order derivative algorithm and universal two-term law). *Let*

$$Q_m = \sum_{n \geq 1} q_{m,n} z^n \quad (m \geq 1), \quad (11.11)$$

with out-of-range coefficients interpreted as zero. Then

$$\begin{aligned} q_{m+1,n} = & -m q_{m,n} + \sum_{r=1}^{n-1} e_r q_{m,n-r} - \frac{1}{L} \partial_\theta q_{m,n-1} \\ & - \frac{1}{L} \{ \partial_\ell q_{m,n-2} - (n-2) q_{m,n-2} \}. \end{aligned} \quad (11.12)$$

This computes the expansion of every fixed $G^{(m)}$ to arbitrary order. If

$$H_r = \sum_{j=1}^r \frac{1}{j}, \quad H_0 = 0, \quad (11.13)$$

then for every fixed $m \geq 1$,

$$Q_m = (-1)^{m-1} (m-1)! z \left[1 + \{e_2 - H_{m-1}\} z + \mathcal{O}_m((1+\ell)z^2) \right]. \quad (11.14)$$

Equivalently,

$$G^{(m)}(y) = (-1)^{m-1} (m-1)! \frac{G(y)}{y^m \rho} \times \left[1 + \frac{(\Psi'(\theta) - 1)/L - H_{m-1}}{\rho} + \mathcal{O}_m\left(\frac{1+\ell}{\rho^2}\right) \right]. \quad (11.15)$$

Proof. Apply equation (11.2) termwise to equation (11.11) and substitute into equation (11.10); coefficient extraction gives equation (11.12). For the two-term formula, write

$$Q_m = a_m z (1 + b_m z + \mathcal{O}_m((1+\ell)z^2)). \quad (11.16)$$

The operator $\mathcal{Y}$ raises the displayed z -order, so through z^2 the recurrence gives

$$a_{m+1} = -m a_m, \quad b_{m+1} = b_m - \frac{1}{m}. \quad (11.17)$$

Starting from $Q_1 = \mathcal{E} = z(1 + e_2 z + \dots)$ yields $a_m = (-1)^{m-1} (m-1)!$ and $b_m = e_2 - H_{m-1}$. $\square$

Corollary 11.3 (Endpoint sign alternation). *For each fixed $m \geq 1$, there exists $y_m > 0$ such that*

$$(-1)^{m-1} G^{(m)}(y) > 0, \quad 0 < y < y_m. \quad (11.18)$$

Thus the inverse Fabius quantile is eventually increasing, concave, with alternating higher derivatives near the left endpoint.

Proof. The bracket in equation (11.15) tends to 1, while $G/(y^m \rho) > 0$. $\square$

Remark 11.4. The sign result is fixed- m : it does not assert a common endpoint interval valid for all derivative orders. Uniformity as $m \rightarrow \infty$ is a separate, substantially harder problem linked to the large-order behavior discussed in section 14.

12 Exact symbolic and numerical verification

The accompanying script `inverse_fabius_asymptotics.py` has two independent verification layers.

12.1 Exact symbolic checks

A generic truncated series $\Phi(z, w)$ is built from algebraically independent symbols

$$L, \beta, C_\sharp, \ell, P_0, \dots, P_4, A_1, A'_1, A_2.$$

The code computes d_1, d_2, d_3 by

1. the triangular substitution $D = z\Phi(z, D)$, and
2. the diagonal extraction equation (6.1).

The differences simplify identically to zero at every tested order. It then verifies equations (7.3), (7.6), (8.13) and (8.17). Because the symbols are independent, these are exact polynomial identities rather than floating-point comparisons.

12.2 Exact dyadic quantiles

Let $a_0 = 1$ and

$$a_n = \frac{1}{2^n - 1} \sum_{k=0}^{n-1} \frac{a_k}{(n - k + 1)!}. \quad (12.1)$$

The dyadic values used in the repository imply

$$F(2^{-n}) = 2^{-n(n-1)/2} a_n. \quad (12.2)$$

All a_n are computed as exact rational numbers. Therefore, at

$$y_n = F(2^{-n}), \quad G(y_n) = 2^{-n}, \quad (12.3)$$

there is no numerical inversion or root-finding error. Only evaluation of the asymptotic formula is numerical.

The periodic function and its first four derivatives are evaluated from [equation \(2.7\)](#) at 100 decimal digits, using twelve positive Fourier modes and their conjugates. The first nonzero Fourier coefficient has magnitude

$$|\widehat{\Psi}(1)| \doteq 1.06271266718 \times 10^{-6}, \quad (12.4)$$

so the phase oscillation is small but easily resolvable at high precision.

n	ρ	G_0	through h_1	through h_2	through h_3
5	6.4398	$-3.83 \cdot 10^{-1}$	$7.89 \cdot 10^{-2}$	$-2.17 \cdot 10^{-2}$	$6.41 \cdot 10^{-3}$
10	12.0804	$-2.57 \cdot 10^{-1}$	$2.69 \cdot 10^{-2}$	$-4.40 \cdot 10^{-3}$	$7.19 \cdot 10^{-4}$
20	22.8244	$-1.62 \cdot 10^{-1}$	$8.76 \cdot 10^{-3}$	$-8.24 \cdot 10^{-4}$	$7.40 \cdot 10^{-5}$
40	43.6518	$-9.66 \cdot 10^{-2}$	$2.72 \cdot 10^{-3}$	$-1.43 \cdot 10^{-4}$	$6.96 \cdot 10^{-6}$
80	84.5410	$-5.53 \cdot 10^{-2}$	$8.05 \cdot 10^{-4}$	$-2.34 \cdot 10^{-5}$	$6.05 \cdot 10^{-7}$
120	125.0820	$-3.94 \cdot 10^{-2}$	$3.88 \cdot 10^{-4}$	$-7.92 \cdot 10^{-6}$	$1.40 \cdot 10^{-7}$
160	165.4729	$-3.08 \cdot 10^{-2}$	$2.30 \cdot 10^{-4}$	$-3.64 \cdot 10^{-6}$	$4.92 \cdot 10^{-8}$

Table 2: Signed relative errors $G^{[r]}(y_n)/2^{-n} - 1$. Here $G^{[0]} = G_0$ and $G^{[r]} = G_0 \exp(\sum_{j=1}^r h_j/\rho^j)$ for $1 \leq r \leq 3$. Notation $a \cdot 10^b$ means $a \times 10^b$.

The new third correction improves the error by a factor of approximately 11.1 at $n = 20$, 38.7 at $n = 80$, and 74.0 at $n = 160$ relative to the two-correction approximation. The monotone improvement with depth is consistent with the fixed-order theorem; the finite- n signs are not asserted to be universal.

12.3 Reproducibility command

From the package directory, run

```
python inverse_fabius_asymptotics.py --output-dir results \
  --precision 100 --modes 12 --depths 5 10 20 40 80 120 160
```

The script writes the exact-check log, constants, CSV table, and both PDF and PNG versions of [figure 1](#). It uses no network access.

13 Equivalent forms and connections across the corpus

13.1 A literal expansion in the small argument

The formulas can be stated without the auxiliary symbol ρ . Put

$$R(T) = \sqrt{\frac{2T}{L}}, \quad \Lambda(T) = \frac{1}{2} \log \frac{2T}{L}, \quad \Theta(T) = R(T) + \beta. \quad (13.1)$$

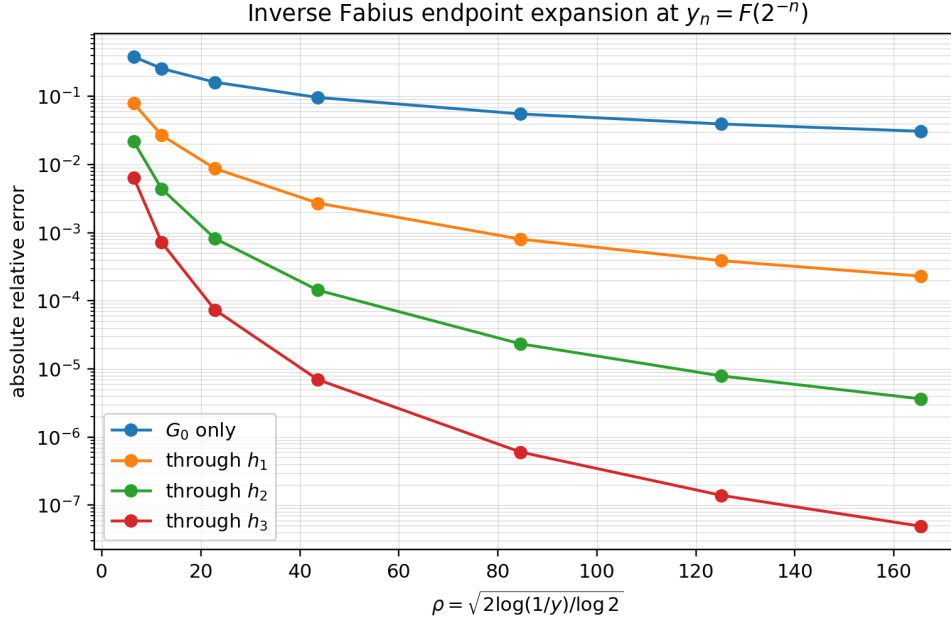


Figure 1: Absolute relative error on the exact dyadic quantiles. Each additional logarithmic correction produces the expected strong reduction once ρ is moderately large.

Then the complete expansion is

$$\begin{aligned}
 G(y) &= \frac{\sqrt{T}}{e\sqrt{L}} \exp(-\sqrt{2LT}) \\
 &\times \exp \left\{ \sum_{n=1}^N \left(\frac{L}{2T} \right)^{n/2} h_n(\Lambda(T), \Theta(T)) \right\} \\
 &\times \left[1 + \mathcal{O}_N \left(\frac{(1 + \log T)^N}{T^{(N+1)/2}} \right) \right], \quad T = \log(1/y) \rightarrow \infty.
 \end{aligned} \tag{13.2}$$

Equivalently, the multiplicative normalization gives

$$\begin{aligned}
 G(y) &= \frac{\sqrt{T}}{e\sqrt{L}} \exp(-\sqrt{2LT}) \left[1 + \sum_{n=1}^N \left(\frac{L}{2T} \right)^{n/2} g_n(\Lambda(T), \Theta(T)) \right. \\
 &\quad \left. + \mathcal{O}_N \left(\frac{(1 + \log T)^{N+1}}{T^{(N+1)/2}} \right) \right].
 \end{aligned} \tag{13.3}$$

The phase $\Theta(T)$ is not an optional decoration: its periodic oscillation is of the same order as the nonoscillatory part of each coefficient.

13.2 Lambert- W geometry

The forward endpoint is naturally parameterized by the exact identity

$$x = \lambda e^{-L\lambda}, \quad \lambda = -\frac{1}{L} W_{-1}(-Lx). \tag{13.4}$$

A direct inversion in x would repeatedly expand W_{-1} and generate nested logarithms. The present method instead performs the probabilistic/saddle inversion in λ and invokes the Lambert identity only once, through [equation \(5.7\)](#). This separation explains why the coefficient algebra is polynomial in $\ell = \log \rho$ rather than an uncontrolled nest of logarithms.

The constant shift $\beta = 1/L + 1/2$ has a geometric role: it completes the square in the dominant quadratic exponent. The leading factor

$$G_0 = \rho e^{-L\rho - L\beta} \quad (13.5)$$

is simply the Lambert map evaluated at the centered saddle phase $\rho + \beta$, with the small reversion D removed.

13.3 Bell and Bernoulli structures

Two distinct Bell-polynomial mechanisms occur.

1. The forward saddle coefficients A_j are differential polynomials assembled from exponential Bell polynomials in the local saddle jets.
2. The inverse multiplicative coefficients g_n are complete Bell polynomials in the logarithmic coefficients h_j , as in [equation \(6.12\)](#).

The diagonal Lagrange–Bürmann formula links these mechanisms without flattening either one. It can therefore be viewed as a “Bell-of-Bell” transform with an additional diagonal grading.

Bernoulli numbers and Bernoulli polynomials enter the forward analysis through Euler–Maclaurin corrections and the expansion of logarithmic factors. On the inverse side, the elementary series

$$\log(1 + \beta z + zD) \quad (13.6)$$

is responsible for the rational terms $-\beta/2$, $-\beta^2/4$, and $\beta^3/3$ seen in [equations \(3.6\)](#), [\(7.3\)](#) and [\(7.6\)](#). At higher orders these elementary contributions can be organized with ordinary Bell polynomials or, after centering, Bernoulli-polynomial translation identities.

13.4 Thue–Morse, q-binomial, and exact dyadic data

The signed Thue–Morse sequence controls finite dyadic products and finite-difference cancellations throughout the corpus. Gaussian q-binomial identities provide exact formulas for dyadic Fabius values and moment recurrences. In the present report their most concrete role is verification: [equations \(12.1\)](#) and [\(12.2\)](#) produce exact rational ordinates y_n whose inverse is known exactly.

There is also a structural analogy. The q-binomial side is graded by dyadic depth, while the inverse expansion is graded by the endpoint scale $z = 1/\rho$. In each case triangular recurrences coexist with closed coefficient extractions. The diagonal formula [equation \(6.1\)](#) suggests searching for a direct q-series packaging of the inverse coefficients, rather than treating the dyadic exact values and continuous endpoint asymptotics as separate subjects.

13.5 The Rvachev sinc product

With the standard normalization, the compactly supported Rvachev function is the symmetric bell

$$\text{up}(x) = F(1 - |x|), \quad |x| \leq 1, \quad (13.7)$$

continued by zero outside its support. Its Fourier transform is an infinite product of sinc factors at dyadic scales. This spectral product explains the simultaneous appearance of dyadic self-similarity, very rapid smoothness, and special exact values.

The inverse endpoint asymptotics describe the geometry of extremely small level sets of the left flank of this bell. Consequently, [equation \(10.2\)](#) may be read as an all-orders formula for the inward displacement from the support boundary at a prescribed tiny height. The Gamma–zeta oscillation is a Mellin shadow of the same dyadic scale lattice that generates the sinc product.

Cross-corpus synthesis

The chain of structures is

$$\begin{aligned} \text{dyadic sinc product} &\longleftrightarrow \text{Thue-Morse/q-products} \longleftrightarrow \text{Mellin lattice and } \Psi, \\ \text{saddle coefficients } A_j &\longrightarrow \text{diagonal inverse coefficients } d_n, h_n, g_n. \end{aligned}$$

The first line and the construction of A_j are developed in the repository. The final diagonal arrow is made closed and explicit in this report.

14 Conjectures and frontier directions

The fixed-order expansion is rigorous. Questions about order $N \rightarrow \infty$, optimal truncation, and complex continuation remain open and appear to contain the most interesting next layer.

14.1 Large-order growth and resurgence

The nearest boundary of the proved common analyticity strip is

$$\sigma_* = \frac{\pi}{2L}. \quad (14.1)$$

The inverse top-jet contribution to h_n combines a derivative of order $2n - 2$ with a divisor $(n - 1)!$. Cauchy's estimate at distance σ_* gives the heuristic scale

$$\frac{(2n)!}{2^n n! L^{2n}} \sigma_*^{-2n} \asymp \frac{n!}{\sqrt{\pi n}} \left(\frac{8}{\pi^2} \right)^n. \quad (14.2)$$

This is factorial, but only Gevrey order one.

Conjecture 14.1 (Gevrey-one inverse expansion). *On every compact real phase set and for fixed bounded ℓ , the coefficient families satisfy bounds of the form*

$$|h_n(\ell, \theta)| + |g_n(\ell, \theta)| \leq C(\ell) A^n n!, \quad (14.3)$$

with the optimal exponential constant controlled by $8/\pi^2$ after the forward normalization is fixed. The Borel transforms have their nearest singularities at locations generated by the complex phase boundaries and their dyadic translates.

The value $8/\pi^2$ is a top-jet heuristic, not yet a proof of the exact large-order constant: other saddle diagrams may reinforce or partially cancel it. A productive route is to combine the forward Bell recurrence with majorant series in a complex strip, then transport the resulting bounds through [equation \(6.1\)](#).

Conjecture 14.2 (Optimal truncation). *For a fixed real phase, truncating near*

$$N_{\text{opt}} \sim \frac{\pi^2}{8} \rho \quad (14.4)$$

produces an exponentially small remainder. Its leading exponential and Stokes multiplier are governed by the nearest complex singularities of Ψ together with the lower-Lambert branch geometry.

14.2 Persistence and geometry of oscillations

Conjecture 14.3 (Nonconstant phase at every order). *For every $n \geq 1$, each of $d_n(\ell, \theta)$, $h_n(\ell, \theta)$, and $g_n(\ell, \theta)$ has a nonconstant periodic component for generic ℓ . Equivalently, no finite truncation beyond the elementary leading factor becomes phase blind.*

The top-jet law makes this highly plausible, but a proof must exclude a special differential identity satisfied by Ψ that cancels all nonzero Fourier modes at some order. A certified proof could use the first Fourier mode, explicit convolution bounds for all nonlinear terms, and the enormous exponential gap between $|k| = 1$ and $|k| \geq 2$.

At first order the cluster set is an interval. At higher order, after removal of logarithmic layers, one obtains images of the circle under vector-valued maps assembled from $\Psi, \Psi', \dots$. Their self-intersections, curvature, and Fourier winding numbers form a new “quantile cluster geometry.”

Conjecture 14.4 (Generic embedded coefficient curves). *For a generic finite selection of renormalized inverse observables, the phase map into $\mathbb{R}^d$ is an immersed analytic closed curve; for sufficiently rich jet selections it is embedded. The minimal embedding dimension is controlled by the first few nonzero Gamma–zeta modes.*

14.3 Certified one-mode and few-mode asymptotics

The first Fourier coefficient has magnitude about 10^{-6} , while the decay exponent is $\pi^2/L \approx 14.24$ per frequency. This suggests a practical theorem: retain the mean and the first conjugate pair in every coefficient, propagate them through the diagonal algebra, and bound the discarded convolution tail by [equation \(9.6\)](#). The output would be a short elementary trigonometric approximation with a rigorous error bar, useful for arbitrary-precision inversion without evaluating many special functions.

A stronger program is to precompute the Fourier-polynomial coefficient tables

$$g_n(\ell, \theta) \approx \sum_{r=0}^n \sum_{|k| \leq K} c_{n,r,k} \ell^r e^{2\pi i k \theta} \quad (14.5)$$

with interval-certified coefficients. Combined with one Newton step on the exact functional equation, this should give a fast globally certified endpoint quantile algorithm.

14.4 General dyadic and non-dyadic smoothing laws

The mechanism is not restricted to the classical Fabius function. Consider smoothing distributions whose Mellin transforms have a scale lattice generated by a base $b > 1$, or compactly supported bells whose Fourier transforms are products

$$\prod_{j \geq 0} \text{sinc}(c_j t) \quad (14.6)$$

with geometric or eventually geometric c_j . Their endpoint logarithms should again have a quadratic main phase, a log-periodic Mellin fluctuation, and differential-polynomial saddle corrections.

Conjecture 14.5 (Universal diagonal inversion class). *Whenever a distribution has a forward exact-phase expansion of the form*

$$-a\lambda^2 + b\lambda - c \log \lambda + C + P(\lambda) + \sum_{j \geq 1} A_j(\lambda) \lambda^{-j}, \quad (14.7)$$

with periodic analytic P, A_j , its inverse admits a diagonal Lagrange–Bürmann expansion with polynomial logarithms, phase-locked subsequences, a newest-jet transfer law, and a Bell-polynomial multiplicative conversion. The universal highest logarithms depend only on a and c .

Proving this abstract theorem would clarify which features are specifically dyadic and which belong to a wider universality class of log-periodically perturbed Gaussian tails.

14.5 Uniform high derivatives and complex quantiles

The recurrence [equation \(11.10\)](#) solves the fixed- m derivative problem. It does not control the double limit $m, \rho \rightarrow \infty$. A two-parameter saddle analysis could identify the transition scale at which the harmonic correction H_{m-1}/ρ ceases to be small and endpoint sign alternation changes character.

Complex continuation is equally promising. The exact phase uses W_{-1} on the real axis, but complex inverse branches interact with the periodic strip singularities. Locating the nearest complex critical values of F would determine the analytic continuation radius of G around tiny positive y and should connect directly to the Borel singularities in [theorems 14.1](#) and [14.2](#).

14.6 Formal verification

The coefficient theorem is well suited to computer-assisted proof. Its core is formal power-series algebra over a differential coefficient ring. A Lean development could proceed in four modules:

1. formal Lagrange–Bürmann inversion with a parameter z ;
2. the diagonal extraction theorem [equation \(6.1\)](#);
3. polynomial-logarithmic degree and top-jet filtrations;
4. analytic transfer from formal cancellation to the fixed-order remainder using [equation \(5.6\)](#).

The accompanying SymPy checks provide test vectors but are not substitutes for such a proof.

15 Conclusion

The inverse Fabius endpoint is governed by three coupled scales:

$$\text{Gaussian magnitude } T, \quad \text{slow logarithm } \ell = \frac{1}{2} \log(2T/L), \quad \text{dyadic phase } \theta = \sqrt{2T/L} + \beta. \quad (15.1)$$

The repository had already identified these scales, proved the all-orders forward expansion, and constructed a recursive inverse continuation. The central advance of this report is to close the coefficient problem: every inverse coefficient is now a finite diagonal Lagrange–Bürmann extraction, with direct companion formulas for both logarithmic and multiplicative normalizations.

This closed calculus reveals structural laws that are difficult to see recursively. The multiplicative highest logarithm is universal, the newest periodic derivative transfers from A_{n-1} with a simple coefficient, every Fourier mode is a finite Gamma–zeta convolution, and every fixed ordinary derivative follows from one differential recurrence. The explicit third correction is already numerically effective on exact dyadic quantiles.

The likely next frontier is not another isolated low-order term. It is the large-order theory: Gevrey bounds, optimal truncation, Borel singularities, and a transseries that resolves the exponentially small remainder. The diagonal formula provides a concrete algebraic foundation for that program.

A Coefficient-extraction proof details

A.1 Why the diagonal formula is finite

For d_n , the summand with index k in [equation \(6.1\)](#) asks for $z^{n-k}w^{k-1}$. Hence:

- no power z^r of Φ with $r > n - k$ contributes;
- no Taylor jet in w above order $k - 1$ contributes;
- an A_j term can occur only when $j \leq n - k$.

Therefore A_{n-1} appears only in the $k = 1$ summand, which is also the key observation in the inverse top-jet proof.

A.2 A partition form

Expand

$$\Phi(z, w) = \sum_{r,s \geq 0} \phi_{r,s} z^r w^s. \quad (\text{A.1})$$

Then [equation \(6.1\)](#) becomes the explicit finite composition sum

$$d_n = \sum_{k=1}^n \frac{1}{k} \sum_{\substack{r_1 + \dots + r_k = n-k \\ s_1 + \dots + s_k = k-1}} \prod_{j=1}^k \phi_{r_j, s_j}. \quad (\text{A.2})$$

This is a general coefficient formula with no implicit lower-order d_j . Grouping equal pairs (r, s) rewrites it as a multivariate ordinary Bell polynomial. Formula [equation \(A.2\)](#) is often preferable for proof, while [equation \(6.1\)](#) is preferable for implementation.

A.3 Formal-to-asymptotic transfer

Let

$$\mathcal{F}_y(w) = \frac{L}{z} w + \mathcal{R}_N(z, w) + E_N(z, w), \quad (\text{A.3})$$

where $\mathcal{R}_N$ contains the retained $A_1, \dots, A_{N-1}$ and $E_N = \mathcal{O}_N(z^N)$. Formal cancellation gives

$$\mathcal{F}_y(D^{[N]}) = \mathcal{O}_N((1 + \ell)^N z^N). \quad (\text{A.4})$$

The exact D satisfies $\mathcal{F}_y(D) = 0$. For small enough z ,

$$\inf_w |\partial_w \mathcal{F}_y(w)| \geq \frac{L}{2z} \quad (\text{A.5})$$

throughout the relevant $\mathcal{O}(\ell z)$ neighborhood. Hence

$$|D - D^{[N]}| \leq \frac{2z}{L} |\mathcal{F}_y(D^{[N]})| = \mathcal{O}_N((1 + \ell)^N z^{N+1}). \quad (\text{A.6})$$

This elementary gain-of-one-power argument is why a forward expansion through A_{N-1} suffices for an inverse phase expansion through d_N .

B Corpus audit ledger

The audit covered every LaTeX source exposed under the requested directory and its subdirectories at the time of access. The tree was actively consolidated on 28 August 2026, so file counts in older internal manifests and the living tree do not coincide. To keep the provenance useful after renames, the ledger below records stable source families and recognizable file stems rather than asserting a permanently frozen count.

Table 3: Stable source families used in the corpus audit.

Family or file stem	Material checked for duplication and used as context
Fabius_Function_and_Rvac hev_Up	Main normalization, functional equations, exact values, moments, Fourier/product identities, endpoint discussion.
FabiusFunction_Mathematical_Glossary	Notation dictionary, theorem crosswalk, constants, special-function conventions.
Non_Elementarity	Differential-algebraic context and restrictions on elementary closed forms.

Family or file stem	Material checked for duplication and used as context
fabius_lean_walkthrough	Formalized definitions and theorem correspondence.
non-formalized-research-frontiers	Exact-phase saddle expansion, coefficient recurrences, Bell-polynomial formulas, top-jet results, open problems.
Inverse_and_Sampling_Frontiers	Leading inverse equivalent, first two corrections, cluster interval, elasticity, all-order triangular recursion, sampling questions.
Thue_Morse_Atlas_and_Frontiers and earlier Thue–Morse drafts	Signed/unsigned sequence formulas, finite products, exact dyadic identities, spectral and q-series links.
Exponents_and_q_Series_Frontiers	Gaussian q-binomial formulas, exponent arithmetic, finite-depth recurrences.
Spectra_and_Arithmetic_Frontiers	Fourier/Mellin spectra, arithmetic values, Dirichlet and Gamma–zeta structures.
Integration_and_Transform_Frontiers; Fabius_Integral_Transforms_Report	Mellin, Laplace, Fourier, resolvent, and integral-transform representations.
Representation_Frontiers; Representation_Second_Wave	Shift/scale systems, polynomial reproduction, multiresolution, moment representations.
Frontier_Compilations	Consolidated reports and cross-topic theorem indexes.
Fourier-decay and sinc-product sources	Partial and infinite sinc products, dyadic spectral decay, Rvachev approximants, numerical Fourier analysis.
Archived small-argument and asymptotic notes	Earlier leading and subleading endpoint formulas, Lambert– W approximations, historical normalizations.
Embedded paper transcriptions	Classical Fabius/Rvachev sources, arithmetic evaluations, smooth compactly supported functions, computational methods.
Interpolation, summation, integration, and representation reports	Screened for inverse expansions, coefficient formulas, and derivative laws so that adjacent new work was not inadvertently repeated.

The immediate novelty comparison was made against the living inverse-focused report, not merely against older archived notes. In particular, the first two corrections and the triangular all-orders engine are explicitly credited as prior repository results throughout this report.

C Reproducibility files and implementation notes

The archive accompanying this report contains:

File	Purpose
<code>inverse_fabius_all_orders.tex</code>	Complete report source.
<code>inverse_fabius_all_orders.pdf</code>	Rendered report.
<code>inverse_fabius_asymptotics.py</code>	Exact dyadic recurrence, Gamma–zeta evaluation, diagonal/triangular symbolic checks, tables, plot.
<code>results/endpoint_errors.csv</code>	Full-precision numerical table underlying table 2 .
<code>results/symbolic_checks.txt</code>	Exact SymPy identity log.
<code>results/constants.txt</code>	Constants and leading Fourier mode at 100 digits.
<code>results/endpoint_error_plot.pdf</code>	Vector version of the error plot.
<code>f</code>	
<code>results/endpoint_error_plot.png</code>	Raster version included in this PDF.
<code>g</code>	

The exact rational recurrence is deliberately separated from the asymptotic evaluator. This prevents numerical evaluation of F or numerical root finding from masking an error in the inverse expansion. The symbolic checker likewise treats the forward jets as independent variables; it therefore checks algebraic reversion rather than only one numerical realization of Ψ .

D Selected notation

Table 4: Notation used throughout the report.

Symbol	Meaning
F, G	Fabius function and its inverse.
L	$\log 2$.
B	$1 + L/2$.
β	$B/L = 1/L + 1/2$.
T	$\log(1/y)$.
ρ	$\sqrt{2T/L}$.
z	$1/\rho$.
ℓ	$\log \rho$.
θ	$\rho + \beta$, interpreted modulo one inside periodic functions.
$\lambda(x)$	$-W_{-1}(-Lx)/L$, so $x = \lambda e^{-L\lambda}$.
Ψ	Mean-zero one-periodic Gamma–zeta endpoint fluctuation.
A_j	Periodic forward saddle coefficients.
D	Phase displacement $\lambda(G(y)) - \rho - \beta$.
d_n	Coefficients of $D = \sum d_n z^n$.
h_n	Coefficients of $\log(G/G_0) = \sum h_n z^n$.
g_n	Coefficients of $G/G_0 = 1 + \sum g_n z^n$.
Φ	Fixed-point kernel in $D = z\Phi(z, D)$.
$\mathcal{E}$	Endpoint elasticity yG'/G .
Q_m	Dimensionless derivative ratio $y^m G^{(m)}/G$.

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